

Assignment 2: Covariance, VaR, and Copulas

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This response answers all five problems in the supplied Assignment 2 PDF. Each problem separates the prediction, model estimates, and reconciliation. Predictions were recorded in `Codes/predictions.md` after examining the permitted descriptive evidence and before calculating model fits or risk measures.

Conventions. Arithmetic returns use the supplied row order. Variance and standard deviation use $n - 1$ unless stated otherwise. The first four moments are mean, sample variance, empirical skewness, and empirical **excess** kurtosis (ordinary kurtosis equals excess kurtosis plus 3). Positions are dollar investment values. A loss has a positive sign:

$$\text{VaR}_\alpha = -q_\alpha(\text{P\&L}), \quad \text{ES}_\alpha = -\text{mean of the worst } \alpha \text{ fraction of P\&L}.$$

Historical VaR uses NumPy's linear quantile. ES averages exactly the worst αn observations, including fractional boundary mass if needed. **A negative VaR represents a gain at the quantile**, rather than a negative loss; these values are retained and explicitly interpreted in Problem 3. Flooring them at zero would change the requested mathematical subadditivity check. Simulations use 100,000 scenarios, fixed seeds, and common normal draws when comparing assumptions.

1. Correlations from mismatched histories

Predict: joint observations and parts (a)-(c)

The table counts each pair's jointly observed days; diagonal entries are each series's available history. All five series trade together on **81 days**.

Series	A	B	C	D	IDX
A	242	242	221	93	242
B	242	242	221	93	242
C	221	221	228	83	228
D	93	93	83	96	96
IDX	242	242	228	96	250

(a) Complete-case correlation is guaranteed PSD when computed on the same nonconstant columns and complete rows: it is a scaled Gram matrix. For a centered, standardized data matrix Z , $R = Z^T Z / (n - 1)$ and $v^T R v = \|Zv\|^2 / (n - 1) \geq 0$ for every vector v . PSD does not by itself guarantee that ordinary Cholesky succeeds, because a singular matrix is PSD but not positive definite. Pairwise correlation can be indefinite. Every pair uses its own subset and means, so the assembled entries need not correspond to any single Gram matrix, even though each entry is individually a valid correlation.

(b) If IDX were exactly the weighted basket, the five-dimensional covariance and correlation matrices would have a zero eigenvalue. Small tracking error makes the true smallest eigenvalue small and positive rather than comfortably far from zero. This sample is therefore particularly exposed to estimation inconsistencies: small entry changes can push that near-zero eigenvalue below zero.

(c) I expect D-related correlations to be least reliable because D has only 96 observations. C-D is the weakest by sample count, at 83 joint days; A-D and B-D have 93 each. The complete-case estimator uses just 81 days even for pairs whose full shared histories are much longer. Counts indicate precision but do not prove that the overlapping periods represent the same underlying regime.

Fit: parts (d)-(f)

(d) The complete-case correlation matrix is:

Series	A	B	C	D	IDX
A	1.000000	0.396728	0.246622	0.413553	0.871358
B	0.396728	1.000000	0.316887	0.236053	0.670860
C	0.246622	0.316887	1.000000	0.185180	0.564842
D	0.413553	0.236053	0.185180	1.000000	0.571930
IDX	0.871358	0.670860	0.564842	0.571930	1.000000

Its eigenvalues in ascending order are:

0.001505787, 0.548345279, 0.693319552, 0.870441029, 2.886388352 .

The pairwise correlation matrix is:

Series	A	B	C	D	IDX
A	1.000000	0.468626	0.430888	0.438449	0.869584
B	0.468626	1.000000	0.463023	0.276692	0.732867
C	0.430888	0.463023	1.000000	0.184615	0.713575
D	0.438449	0.276692	0.184615	1.000000	0.605691
IDX	0.869584	0.732867	0.713575	0.605691	1.000000

Its eigenvalues in ascending order are:

-0.011071847, 0.471421960, 0.528366787, 0.857381135, 3.153901965 .

Ordinary `numpy.linalg.cholesky` succeeds for complete-case correlation and fails for pairwise correlation. The negative pairwise eigenvalue is materially below floating-point error, so the failure is substantive.

(e) I construct $\Sigma = D_s R_{pair} D_s$ with D_s holding each series's full-history sample standard deviation. These are:

Series	Std
A	0.01650183
B	0.01191893
C	0.01832355
D	0.02282866
IDX	0.01228141

The tracking portfolio has weights $w = (-0.40, -0.30, -0.20, -0.10, 1)^T$ in the order A, B, C, D, IDX. Its variance $w^T \Sigma w$ is **-1.799369655e-06**, in squared return units.

(f) Rebonato-Jackel clips negative correlation eigenvalues to zero and rescales the diagonal to one. Higham alternates projections onto the PSD cone and unit diagonal with Dykstra's correction. The repository implements both. I use Higham tolerance 10^{-12} , retain full-history standard deviations, and recompute the same tracking variance.

Method	Smallest eigenvalue	Frobenius distance	Tracking variance
Complete case	0.001506	0.423935	1.097545e-06
Pairwise	-0.011072	0	-1.799370e-06
Rebonato-Jackel	-7.515355e-17	0.016182	6.807298e-07
Higham	-8.604428e-13	0.015520	6.787649e-07

The repaired eigenvalues near zero are numerical zeros; Higham's tiny negative value is within its stated tolerance. Both repairs restore a nonnegative tracking variance. A repaired correlation can be singular, so ordinary Cholesky need not succeed for it; PSD simulation requires a factorization that permits zero pivots. The complete-case row in this table also uses the full-history standard deviations to isolate the change in correlation estimator.

Reconcile: parts (g)-(i)

(g) PSD means $w^T \Sigma w \geq 0$ for **every** portfolio vector w . The negative variance in (e) directly violates this definition. It is not evidence that this hedge is safer than zero risk: it demonstrates that the matrix is not a valid covariance matrix. Congruence with positive D_s preserves the failure of PSD, and this particular near-basket direction exposes it.

(h) Higham's largest three entry changes are shown first below. Symmetric entries have equal changes and are counted once.

Pair	Overlap	Pairwise	Higham	Change
A-IDX	242.000000	0.869584	0.862540	-0.007045
C-IDX	228.000000	0.713575	0.708810	-0.004765
B-IDX	242.000000	0.732867	0.728687	-0.004180
D-IDX	96.000000	0.605691	0.601964	-0.003727
A-C	221.000000	0.430888	0.433165	0.002277
A-B	242.000000	0.468626	0.470623	0.001997
A-D	93.000000	0.438449	0.440230	0.001781
B-C	221.000000	0.463023	0.464374	0.001351
C-D	83.000000	0.184615	0.185820	0.001205
B-D	93.000000	0.276692	0.277749	0.001057

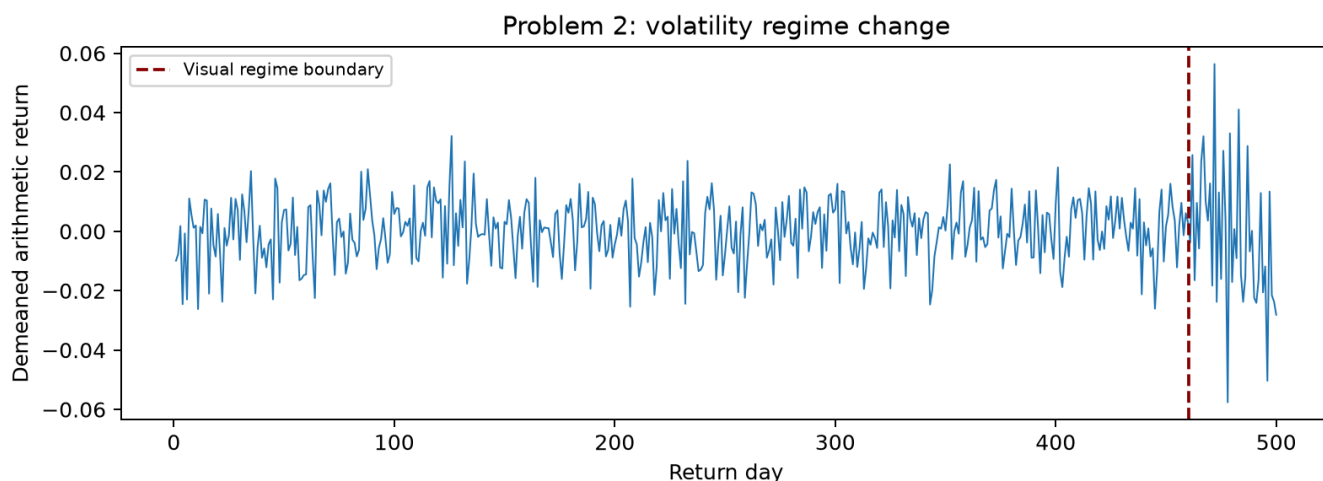
The largest movements involve IDX, especially A-IDX, rather than the least observed C-D pair. That differs from my reliability ranking but is expected from the algorithm's objective. Unweighted Higham minimizes Frobenius distance subject to PSD and a unit diagonal; it uses the geometry of the inconsistent eigenstructure, not observation counts or entry-specific estimation uncertainty. The near-linear index relationship drives the constraint violation. A reliability-weighted repair would pose a different optimization problem.

(i) The complete-case/pairwise gap is **0.423935** in Frobenius norm, compared with **0.016182** for Rebonato-Jackel and **0.015520** for Higham. The two repairs differ by only **0.004579**. Here the estimator choice changes the matrix much more than either the repair's magnitude or the choice between repairs. Repair is still necessary to make the pairwise matrix admissible; numerical admissibility alone does not settle which sampling estimator is more reliable.

2. A volatility estimate after the regime changed

Predict: parts (a)-(c)

I compute 500 arithmetic returns from the 501 prices and remove their full-sample mean, **0.000161876**. The plot suggests a change near day 461, so I identify the most recent 40 returns as a visibly higher-volatility period. This is a visual split, not an estimated or assumed known change point.



(a) My recorded smallest-to-largest prediction was:

fitted t < historical < equal-weight normal < EW normal 0.97 < EW normal 0.94.

The recent high-volatility days should dominate EW estimates, especially with 0.94. A full-sample normal standard deviation is inflated by the volatile period. I expected a t fit to emphasize the quieter center while allowing large shocks; the historical-versus-t-versus-normal ordering was less certain because the 5% quantile depends on observed lower-tail frequency, not kurtosis alone.

(b) For normalized weights $w_j \propto \lambda^j$, with age $j = 0$ for the newest return, the finite-sample effective sample size is $n_{eff} = 1 / \sum w_j^2$. Its infinite-history limit is $(1 + \lambda) / (1 - \lambda)$. Half life is $\log(0.5) / \log \lambda$. The last $m = 40$ days receive $(1 - \lambda^m) / (1 - \lambda^{500})$ of total weight.

Lambda	Effective sample	Half life	Recent 40 weight
0.970000	65.666635	22.756573	0.704288
0.940000	32.333333	11.202306	0.915838

Thus 0.94 places about 92% of its weight on the visible new period, whereas 0.97 places about 70% there. They are estimating different mixtures of old and new volatility, not simply applying two equally precise estimates to the same regime.

(c) The full-sample moments are:

Series	Mean	Variance	Skewness	Excess kurtosis
Return	-2.220446e-19	1.418981e-04	-0.155587	2.132367

Excess kurtosis **2.132367** is large relative to the normal benchmark of zero, while skewness is modest. A rough normal sampling standard error for excess kurtosis is $\sqrt{24/500} \approx 0.219$; this is a descriptive benchmark, not a definitive test for these potentially nonstationary returns. The plot and moments make a single constant-volatility normal regime implausible as a useful description. A mixture of normal regimes can produce heavy unconditional tails without heavy tails within each regime.

Fit: part (d)

All VaRs below are one-day dollar values for a USD 1,000,000 position, at 5%. Normal estimates use zero mean after full-sample demeaning. EW variance follows the existing library convention: a weighted local mean and normalized weights, without an extra $n - 1$ correction. A constant shift of the series leaves that weighted variance unchanged.

Method	VaR
Equal-weight normal	19,593.66
EW normal 0.97	34,063.74
EW normal 0.94	36,307.95
Fitted t	18,872.44
Historical	20,514.50

The fitted t parameters are location **0.000181639**, scale **0.010289990**, and degrees of freedom **8.269918**. Its scale is not its standard deviation. Although the sample was demeaned, an unrestricted t likelihood can have a nonzero fitted location; I retain it when computing $-\mu - s t_{\nu}^{-1}(0.05)$.

Reconcile: parts (e)-(g)

(e) The actual ranking is:

Fitted t < Equal-weight normal < Historical < EW normal 0.97 < EW normal 0.94.

My recent-weighted and t predictions were supported, but historical and equal-weight normal reversed. I had treated overall variance inflation as a sufficient guide to the empirical 5% quantile. The lower-tail observations put that quantile farther out than the normal's 1.645-standard-deviation threshold. Heavy unconditional tails do not force every quantile to exceed the normal value: the fitted t has a smaller central scale and the lowest 5% VaR in this sample.

(f) Using the same visually selected boundary:

Regime	Days	Mean	Std
Days 1-460	460.000000	0.000209	0.010178
Days 461-500	40.000000	-0.002406	0.024289

The newer period has roughly 2.4 times the older standard deviation. For a zero-mean mixture of normal regimes, $\kappa_{excess} = 3E[\sigma^4]/E[\sigma^2]^2 - 3$. Inserting the regime sample variances and fractions 460/500 and 40/500 gives **2.572275**, the same order as the observed excess kurtosis. This illustration neglects the small regime mean differences and finite-sample fourth-moment error. It explains why pooled returns can look t-like: the fitted t describes an **unconditional variance mixture**, rather than proving that tomorrow's shocks follow one stable t law. This observation alone does not establish normality within each regime.

(g) Applying the supplied approximation $SE(\hat{\sigma}) \approx \hat{\sigma} / \sqrt{2n_{eff}}$:

Lambda	Sigma	Sigma SE	VaR SE
0.970000	0.020709	0.001807	2972.384575
0.940000	0.022074	0.002745	4515.039136

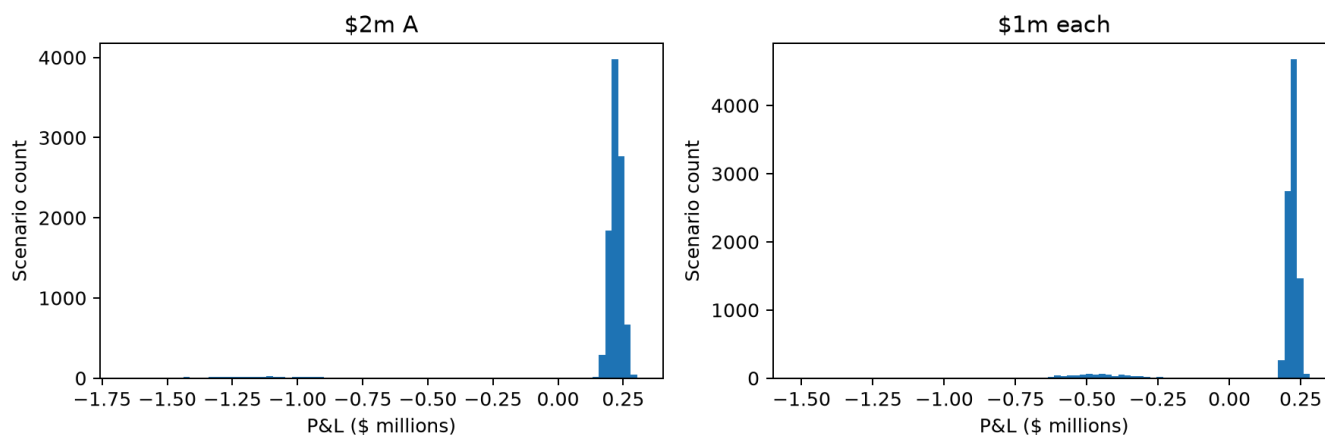
The EW VaR gap is **USD 2,244.21**, smaller than either individual approximate VaR standard error. It is therefore not clearly larger than the stated estimation noise benchmark. The two estimates use the same observations and are correlated; their individual standard errors do not provide the standard error of their difference, and adding them as if independent would not be a formal test. The normal-regime approximation is also imperfect during a volatility transition.

Opinion: I would use 0.94 for a trading limit because it reacts faster to current market risk, while monitoring the instability caused by its effective sample of only about 32 days. I would prefer 0.97 among these two for a capital number because its longer memory is more stable. Capital also needs stress scenarios and a specified horizon; the slower decay factor alone does not make this one-day estimate an adequate capital methodology.

3. When diversification raises VaR

Predict: parts (a)-(b)

The two histograms contain a large repayment cluster and a small default-loss cluster. Splitting the investment usually reduces the loss when only one bond defaults, but it creates more scenarios with some loss.



(a) Bond-return moments are:

Series	Mean	Variance	Skewness	Excess kurtosis
A	0.083709	0.018493	-4.822523	22.081997
B	0.085011	0.017451	-4.914778	23.074367

Both bonds are strongly negatively skewed and have excess kurtosis above 22. A symmetric normal approximation spreads mass smoothly between the repayment and default outcomes, whereas these scenarios have separated clusters. Mean and variance therefore cannot locate the default threshold adequately. A normal VaR can miss both

the sign of the individual 5% quantile and the portfolio's change in default probability.

(b) Losses exceeding 20% occur in **406 A scenarios (4.06%)**, **393 B scenarios (3.93%)**, and **782 union scenarios (7.82%)**. Both default in **17 scenarios (0.17%)**. My prediction was that diversification raises 5% VaR but lowers ES relative to investing all USD 2m in A. For concentrated A, the 5th percentile lies beyond its default cluster in the repayment outcomes. For the diversified choice, 7.82% of scenarios have at least one default, so its 5th percentile falls within a loss cluster. ES should improve because most of these defaults affect only half the investment. Counts alone do not determine ES; the severity pattern visible in the P&L plots supports that prediction.

Fit: parts (c)-(e)

(c) Historical 5% risk, in dollars:

Position	VaR 5%	ES 5%
USD 1m A	-84,569.80	443,628.30
USD 1m B	-86,063.46	419,587.29
USD 2m A	-169,139.60	887,256.60
USD 1m each	414,604.15	541,997.24

For the individual positions and concentrated A, the negative 5% VaR is a **positive gain at the lower quantile**. For example, the USD 2m A position's 5th percentile is a gain of USD 169,139.60. Its ES is nevertheless a large positive loss because ES includes all the severe defaults in the worst 5%. Negative VaR does not mean defaults are absent.

(d) Normal 5% VaR, using each position's sample P&L mean and standard deviation:

Position	Normal VaR 5%
USD 1m A	139,970.62
USD 1m B	132,276.77
USD 2m A	279,941.24
USD 1m each	143,466.25

Here $\text{VaR} = -\bar{P} + 1.644854 s_P$. I retain the positive expected bond repayment in $\bar{P}$ rather than incorrectly imposing zero mean.

(e) Historical 1% risk, in dollars:

Position	VaR 1%	ES 1%
USD 1m A	649,701.27	709,547.78
USD 1m B	640,392.63	700,805.34
USD 2m A	1,299,402.55	1,419,095.55
USD 1m each	598,122.57	716,407.69

Reconcile: parts (f)-(h)

(f) For the two USD 1m positions, 5% VaR subadditivity would require $\text{VaR}(A + B) \leq \text{VaR}(A) + \text{VaR}(B)$. Numerically the left side is **USD 414,604.15** and the right side is **USD -170,633.26**. The inequality fails. Individual 5% quantiles hide defaults whose probabilities sit below 5%; the union pushes the combined position's loss probability above that cutoff. This is a quantile threshold effect, not a claim that diversification increases every notion of risk.

ES satisfies the same check: the combined ES is **USD 541,997.24**, below the sum **USD 863,215.59**. It is also below the concentrated USD 2m A ES of **USD 887,256.60**. Averaging the entire tail captures loss severity and rewards spreading most defaults over half the capital. Both parts of my prediction are supported.

(g) Normal VaR favors USD 1m in each: its value is USD 143,466.25 versus USD 279,941.24 for USD 2m A. This is reasonable if the economic criterion is lower severe-loss exposure, as ES also favors the split. It is **not correct for the requested historical 5% VaR ranking**. The normal reaches its conclusion through lower variance and a smooth symmetric distribution, which does not describe the probability mass crossing the 5% threshold. Thus it does not identify the actual mechanism in these scenarios.

(h) At 1%, each individual VaR is inside the default-loss cluster. The diversified VaR is now smaller than concentrated A's VaR, and the subadditivity inequality holds for this data: the combined value is USD 598,122.57, compared with the sum USD 1,290,093.90. The 0.17% simultaneous-default frequency is below 1%; most of the worst 1% still consists of single defaults. VaR's failure depends on the probability level and the shape and placement of probability mass. Its success at 1% here is not a general proof of coherence.

4. Gaussian or t copula

Predict: parts (a)-(b)

(a) The return moments are:

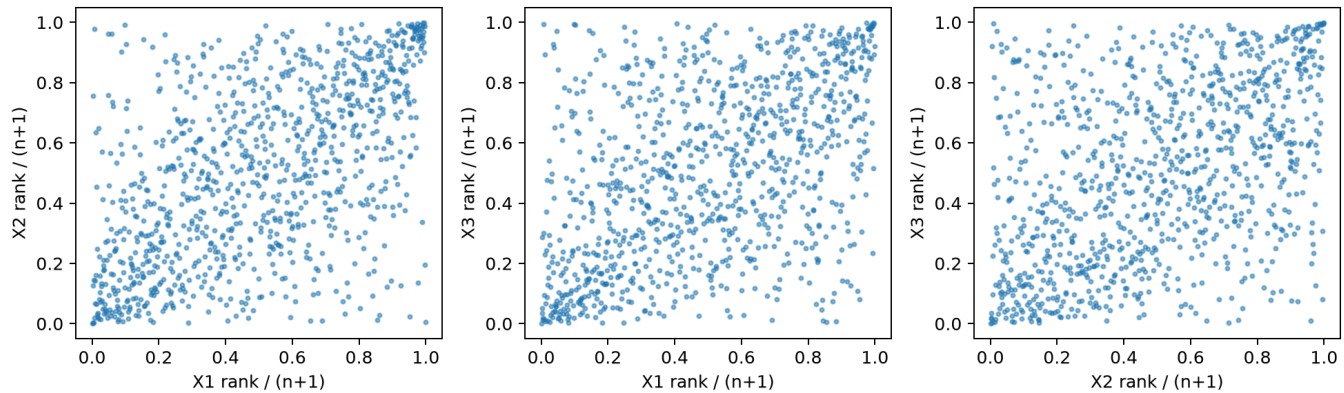
Series	Mean	Variance	Skewness	Excess kurtosis
X1	2.565850e-04	1.414697e-04	-0.034174	-0.122542
X2	-2.523809e-04	2.614199e-04	-0.614117	7.083388
X3	8.897999e-04	1.003768e-04	0.073955	2.853745

I predict a normal margin for X1 and t margins for X2 and X3. X1 is nearly symmetric with near-zero excess kurtosis. X2 and X3 look heavier-tailed. X2's skewness and particularly its excess kurtosis deserve a sensitivity check. Removing each series's largest absolute deviation from its mean gives:

Series	Row	Value	Skew without	Excess without
X1	952.000000	-0.040000	-0.005487	-0.192699
X2	952.000000	-0.142914	0.055783	1.710093
X3	952.000000	-0.061244	0.312681	1.716058

All three extreme observations occur on row 952. In X2, removing that row changes skewness from about -0.614 to +0.056 and excess kurtosis from about 7.083 to 1.710. It strongly influences the moments, but substantial excess kurtosis remains. A symmetric t accommodates thickness, not genuine asymmetry; one influential observation alone is weak evidence for a stable skewed distribution. I keep the observation in all fits and risk calculations.

(b) I use ranks divided by $n + 1$, so every pseudo-observation lies strictly inside $(0, 1)$. The rank plots show clusters in joint corners, as well as positive association through the center.



The counts of days jointly in the worst or best 2.5% are:

Pair	Lower	Upper
X1-X2	6	8
X1-X3	7	12
X2-X3	6	9

Under independent series the expected count on **each side** is $1000(0.025)^2 = 0.625$ days. Actual counts of 6-12 are much higher. My prediction is that the t copula wins because it can model stronger simultaneous extreme ranks. Independence is only a baseline: positive Gaussian dependence itself also raises joint-tail counts, so exceeding 0.625 is not sufficient to distinguish Gaussian and t copulas. The visual corner concentration motivates the comparison but the fits must decide whether additional tail dependence is justified.

Fit: parts (c)-(f)

(c) Normal and t fits use maximum likelihood. The repository's normal fit returns the sample standard deviation; I multiply it by $\sqrt{(n-1)/n}$ to obtain the normal MLE standard deviation before scoring. The t scale comes directly from the repository's MLE fit. AICc is

$$\text{AICc} = 2k - 2\ell + \frac{2k(k+1)}{n-k-1}.$$

Here normal has $k = 2$ and t has $k = 3$. Smaller scores are preferred.

Margin	Model	Mu	Scale	Nu	Log L	AICc
X1	Normal	2.565850e-04	0.011888	-	3013.274274	-6022.536513
X1	t	2.565563e-04	0.011888	9.884359e+08	3013.274274	-6020.524452
X2	Normal	-2.523809e-04	0.016160	-	2706.253036	-5408.494036
X2	t	-1.759779e-04	0.012544	5.271506	2762.104428	-5518.184759
X3	Normal	8.897999e-04	0.010014	-	3184.851638	-6365.691239
X3	t	7.455853e-04	0.008181	6.094916	3217.019807	-6428.015517

AICc selects **normal for X1, t for X2, and t for X3**, agreeing with my prediction. X1's enormous fitted t degrees of freedom is the normal limit, not convincing evidence for a distinct finite-t tail. Its log likelihood is essentially the normal likelihood and its additional parameter incurs a penalty. AICc is used as the requested comparison; near this boundary its usual regular-model interpretation deserves caution. X2 and X3 show clear finite-t improvements.

(d) I transform returns through each selected fitted marginal CDF. For each pair I estimate Kendall's τ from the returns and set $R_{ij} = \sin(\pi\tau_{ij}/2)$. The resulting matrix is:

Series	X1	X2	X3
X1	1.000000	0.583263	0.502851
X2	0.583263	1.000000	0.464383
X3	0.502851	0.464383	1.000000

Its smallest eigenvalue is **0.412792**, so no PSD repair is needed. I hold this R fixed for both copulas. For Gaussian scores $z_i = \Phi^{-1}(u_i)$, and for t scores $x_i = t_\nu^{-1}(u_i)$, the log copula densities are

$$\log c_G(u) = \log \phi_R(z) - \sum_i \log \phi(z_i), \quad \log c_t(u) = \log f_{t,\nu,R}(x) - \sum_i \log f_{t,\nu}(x_i).$$

The t copula's nu is fitted by a coarse search followed by scalar likelihood optimization. It is a copula parameter independent of either t margin's nu. CDF values are clipped only at 10^{-12} and $1 - 10^{-12}$ to keep inverse CDFs finite.

Copula	Nu	k	Log L	AICc	BIC
Gaussian	infinity	3.000000	350.440286	-694.856477	-680.157307
t	3.659932	4.000000	451.999415	-895.958628	-876.367808

I count the three estimated off-diagonal R entries for Gaussian, and those three plus nu for t. Both scores exclude the common marginal parameters. BIC is $k \log(n) - 2\ell$. This is the specified two-stage, Kendall-plug-in comparison, not a joint maximum-likelihood estimate of R and the margins. If one instead conditions on R and

counts only the subsequently optimized parameter, the absolute criterion values change but the strong preference here remains.

(e) For each copula, I generate 100,000 correlated normal scores with the repository simulator. Gaussian uniforms are $\Phi(Z_i)$. For t, each scenario uses a **single shared** chi-square scale: $T_i = Z_i/\sqrt{W/\nu}$, $W \sim \chi_\nu^2$ independent of Z, followed by $u_i = t_\nu(T_i)$. Independent chi-square scales per asset would produce the wrong copula. I then apply the selected marginal inverse CDFs and sum the P&L from USD 1m in each asset. Common Z draws reduce simulation noise in the comparison.

Model	VaR 5%	ES 5%	VaR 1%	ES 1%
Gaussian	49,410.42	66,382.95	76,233.73	93,826.92
t	48,828.51	68,097.06	79,559.93	100,316.65
Historical	47,541.34	65,698.80	67,264.25	102,424.22

All values are dollars. Historical estimates use the 1,000 observed portfolio P&Ls. At 1%, historical ES is based on only 10 days, whereas each simulation has 1,000 days in its worst 1%; differences include sampling uncertainty and distributional assumptions.

(f) Implied joint-tail days per 1,000 days are simulation frequencies multiplied by 1,000. Thresholds are the simulated margins' population 2.5%/97.5% uniforms, matching the tail fractions in the rank-count exercise.

Model	Pair	Lower	Upper
Gaussian	X1-X2	6.39	6.38
Gaussian	X1-X3	4.61	4.68
Gaussian	X2-X3	4.55	4.49
t	X1-X2	9.46	9.44
t	X1-X3	8.11	8.20
t	X2-X3	7.82	8.00

The two elliptical copulas imply symmetric lower and upper dependence. Small lower/upper differences in the simulated values come from Monte Carlo error.

Reconcile: parts (g)-(j)

(g) The t copula wins: its log likelihood is higher by **101.559128**, AICc is lower by **201.102152**, and BIC is lower by **196.210501**. The fitted nu is **3.659932**, consistent with meaningful tail dependence. The t simulation better represents the approximately 9.5 average upper/lower count for X1-X3 and 7.5 for X2-X3 than the Gaussian's counts near 4.5. For X1-X2, however, the empirical average is 7 and Gaussian counts around 6.4 are closer than t counts around 9.5. Thus t does not match every pair and side more closely. With 25 observations in each marginal 2.5% tail, the observed lower/upper asymmetry is also noisy and cannot be fully matched by either symmetric copula.

(h) Changes on switching from Gaussian to t are:

Measure	t minus Gaussian	Percent change
VaR 5%	-581.90	-1.18%
ES 5%	1,714.11	2.58%
VaR 1%	3,326.20	4.36%
ES 1%	6,489.72	6.92%

1% ES moves most, increasing by USD 6,489.72. It averages very extreme losses and is strongly affected by simultaneous extreme uniforms. At 5% VaR, the relevant quantile is closer to the body; the margins and R are the same under both models. Redistributing probability into very extreme joint losses can greatly change ES while barely moving that less extreme quantile. For fixed R, a t copula need not increase every VaR level; here 5% VaR decreases slightly.

(i) I calculate every day's contribution $\Delta \ell_d = \log c_t(u_d) - \log c_G(u_d)$ and export all 1,000 values to `results/p4_daily_likelihood.csv`. Defining a non-tail day by **all three fitted uniforms lying between 0.05 and 0.95**, there are **793 days**. They contribute **50.477953** of the total **101.559128**, or **49.70%**. The remaining tail-containing days contribute **51.081175**.

If "outer 5%" is instead defined by empirical ranks, there are **784 central days**, contributing **49.800343** (49.04% of the total). Both definitions give the same substantive conclusion: about half of the likelihood improvement comes from days without an extreme margin. Likelihood scores density over the entire joint distribution. It is not a loss-tail-only score, a count of joint extremes, or a direct optimization of portfolio VaR/ES. A shared-scale t copula also changes central clustering and off-diagonal joint density, which helps explain its broad likelihood advantage.

(j) The most correlated pair is **X1-X2**, with $\rho = 0.583263$. Under a t copula, both asymptotic lower and upper tail-dependence coefficients are

$$\lambda_L = \lambda_U = 2t_{\nu+1} \left(-\sqrt{\frac{(\nu+1)(1-\rho)}{1+\rho}} \right) = 0.321942.$$

For a Gaussian copula with $|\rho| < 1$, both are **zero**. Zero asymptotic tail dependence does not imply independent 2.5% tail events: a positively correlated Gaussian copula can still produce several joint tail days at a finite threshold.

5. Model-based simulation and residual correlation

Predict: part (a)

With $r_i = \alpha_i + \beta_i r_M + \epsilon_i$, residual covariance adds $2V_A V_B \text{Cov}(\epsilon_A, \epsilon_B)$ to portfolio variance. For P1 the exposures have the same sign, so positive residual correlation increases variance; treating residuals as independent understates VaR. For P2 the exposures have opposite signs, so positive residual correlation reduces variance; independence overstates VaR. I predict greater proportional exposure in P2 if the two market betas are similar, because market effects cancel in the spread and residual assumptions account for a larger share of its total risk.

Fit: parts (b)-(d)

(b) I compute 500 arithmetic returns and fit each OLS regression with an intercept. The estimates are:

Series	Alpha	Beta	Residual std (n-1)	Residual std (n-2)
A	2.050126e-04	1.035278	0.012888	0.012901
B	3.147726e-04	0.935417	0.011538	0.011550

The sample residual correlation is **0.698307**. The full residual covariance matrix is:

Series	A	B
A	1.661083e-04	1.038459e-04
B	1.038459e-04	1.331356e-04

Market return standard deviation is **0.010552717**. For the covariance decomposition and simulation I use residual covariances with divisor $n - 1$, consistent with the stock and market sample covariances. I also report the conventional regression residual standard errors using $n - 2$ in the table; substituting those would change the exact sample identity slightly.

(c) I simulate 100,000 normal market/error vectors. Market and residuals are independent in the joint-normal model, with market variance and the 2x2 residual covariance in separate blocks. In the shortcut I retain the residual variances and set only their cross-covariance to zero. The same seeds give common underlying normal draws for the two cases. Simulated stock returns are $r_i^* = \hat{\beta}_i r_M^* + \epsilon_i^*$.

The fitted intercepts are reported in (b), but omitted from simulation because the problem expressly assumes **zero expected returns**. I also set the expected market and error returns to zero. All results below are one-day 5% dollar VaRs.

Assumption	Portfolio	Simulated VaR	Analytical model VaR
Full residual covariance	P1	50,219.93	50,414.62
Full residual covariance	P2	15,792.50	15,833.60
Independent residuals	P1	44,412.93	44,493.99
Independent residuals	P2	28,491.05	28,506.53

(d) Direct delta-normal VaR uses $1.644854 \sqrt{w^T \hat{\Sigma}_{stocks} w}$ with signed dollar weights and zero mean. It is **USD 50,414.62 for P1** and **USD 15,833.60 for P2**. The sample stock covariance matrix used is:

Series	A	B
A	2.854638e-04	2.116886e-04
B	2.116886e-04	2.305761e-04

Reconcile: parts (e)-(f)

(e) The shortcut moves risk in the predicted directions. Using analytical model values to remove simulation noise, it reduces P1 VaR by **USD 5,920.63** (11.74%), and increases P2 VaR by **USD 12,672.92** (80.04%). P2 is more exposed in both proportional VaR terms and the absolute VaR difference. The betas are close, so P2's market coefficient is their difference while P1's is their sum. P2 benefits from positively correlated industry residuals canceling in the spread; independence discards that hedge. At the variance level the absolute change is equal and opposite across the two equally sized portfolios, but the square root converts it to a larger relative VaR change for P2.

(f) In finite-sample OLS with an intercept, the residual means are zero and each residual is orthogonal to the market regressor. With the same $n - 1$ convention,

$$\hat{\Sigma}_{stocks} = \hat{\beta}\hat{\beta}^T\hat{\sigma}_M^2 + \hat{\Sigma}_\epsilon.$$

The largest entrywise numerical discrepancy in this identity is **1.084e-19**. Thus the full residual model's analytical VaR equals the direct delta-normal VaR. Simulated full-model VaR differs by **USD -194.69 for P1** and **USD -41.11 for P2**, small differences due to Monte Carlo quantile noise.

The covariance identity is OLS algebra and does not itself need normality. Joint normality adds the distributional justification: uncorrelated market and errors are independent, and every linear portfolio has a normal distribution. Then its mean and variance determine its VaR. They would stop agreeing as actual risk models if returns have nonnormal residuals, tail dependence, time-varying betas/volatility, or nonlinear exposures and revaluation. The sample identity can also fail if covariances use different histories, incompatible divisors, regularized betas, or omitted cross terms. If both methods continue to impose the same normal covariance approximation, their analytical numbers can still agree even when that common approximation is inaccurate.

Code, reusable library functions, and verification

The submission's main computation is `Codes/run_assignment.py`. The copula likelihood and shared-scale simulation logic are in `Codes/copulas.py`. The tables and figures in this response come from the computed results. No original assignment data or existing library implementation was changed.

The following existing functions are imported and called directly from `FunctionalTest/testfiles/lib` in this repository. These local source modules are the implementation references for the reused calculations:

Library source	Functions actually used	Assignment use
<code>corr_and_covar.py</code>	<code>correlation_missing_data_skip_missing_rows</code> , <code>correlation_missing_data_pairwise</code> , <code>covariance_missing_data_skip_missing_rows</code>	P1 correlations; P5 covariance matrices
<code>fix_nonpsd_corr_covar.py</code>	<code>near_psd</code> , <code>higham_nearest_psd</code>	P1 Rebonato-Jackel and Higham repair
<code>expo_weighted_corr_and_covar.py</code>	<code>ew_covariance</code>	P2 EW volatility
<code>returns_conversion.py</code>	<code>return_calculate</code>	P2/P5 arithmetic returns
<code>distribution_fitting.py</code>	<code>fit_normal</code> , <code>fit_general_t</code> , <code>aicc</code>	P2/P3/P4 fitting and scoring
<code>risk_measures.py</code>	<code>normal_risk</code> , <code>t_risk</code> , <code>sample_risk</code>	Analytical and empirical VaR/ES
<code>simulations.py</code>	<code>simulate_normal</code>	P4 copula scores; P5 factor simulation

`simulate_normal` internally uses `chol_psd` from `chol_psd.py`. I did not use `portfolio_risk` as a substitute for P4: its fixed Gaussian/Spearman convention does not implement the assignment's Kendall-based Gaussian-versus-t comparison. I also did not use `fit_regression_t` for P5, which expressly asks for normal errors.

Run `bash Codes/run_all.sh` from the Assignment2 folder to reproduce all numbers, plots, and mathematical validation. The final `Ans.md` and `Ans.pdf` are included separately as written responses. The README gives environment setup and commands from the repository root as well as this folder.

The relevant existing library test suites passed **165 tests**. Assignment validation separately checks the manual multivariate-t density ratio, the Gaussian limit, simulated uniform margins, Kendall dependence, repaired PSD matrices, ES subadditivity, positive scaling, OLS covariance decomposition, and the aggregation of daily copula likelihood contributions. These checks target mathematical properties rather than memorized numerical answers.

Reproducibility limits: pseudo-random seeds fix the draws in the same numerical environment. Different SciPy optimizers or versions can produce slightly different MLE parameters, especially X1's effectively infinite t nu. Simulation quantiles can therefore vary slightly across environments. The conclusions depend on the reported mechanisms and numerical scale, not exact last-digit matches.

Sources: the assignment requirements and formulas specified in `Assignment 2.pdf`; the supplied `problem1.csv` through `problem5.csv`; and the repository library source files listed above. The pre-fit predictions are preserved separately in `Codes/predictions.md` for review.